

Smart OPD Queue & Patient History Management

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Understanding the Challenge

Analyzing the Bottlenecks in Modern Healthcare Facilities

The Problem Statement



Manual Queue Systems

Reliance on manual paper tokens leads to long, unpredictable waiting times and overcrowded OPD halls.



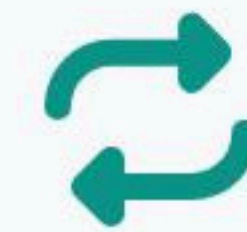
Fragmented Records

Paper-based medical records cause a lack of continuity in patient care and increased clinical risks.



Emergency Gaps

Absence of immediate access to critical medical history during accidents can lead to fatal delays.



Redundant Testing

Inaccessible history results in repeated clinical tests, increasing the financial burden on patients.

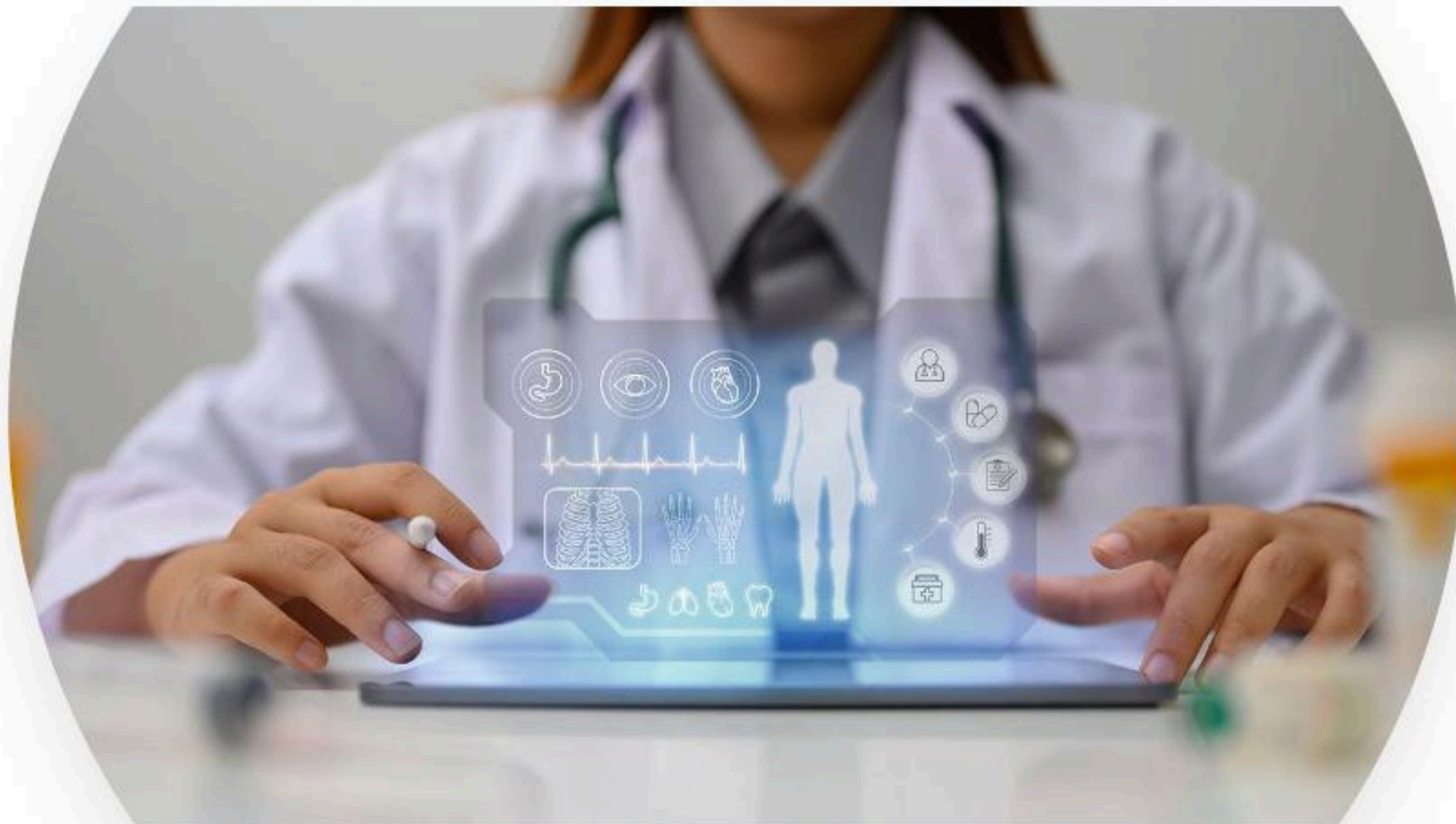
Introduction & Project Context

Modern clinics face significant hurdles in patient flow and data management. Our system addresses these by:

- ✓ Optimizing OPD flow through real-time visibility.
- ✓ Maintaining longitudinal digital medical records.
- ✓ Enhancing clinical decision-making for doctors.



The Core Aim of Our Project



Seamless Workflow Integration

Our primary goal is to develop an integrated system that reduces patient waiting time and improves workflow efficiency.

By providing doctors with instant access to past prescriptions and medical records, we enable better clinical outcomes and faster treatment cycles.

Key Project Objectives

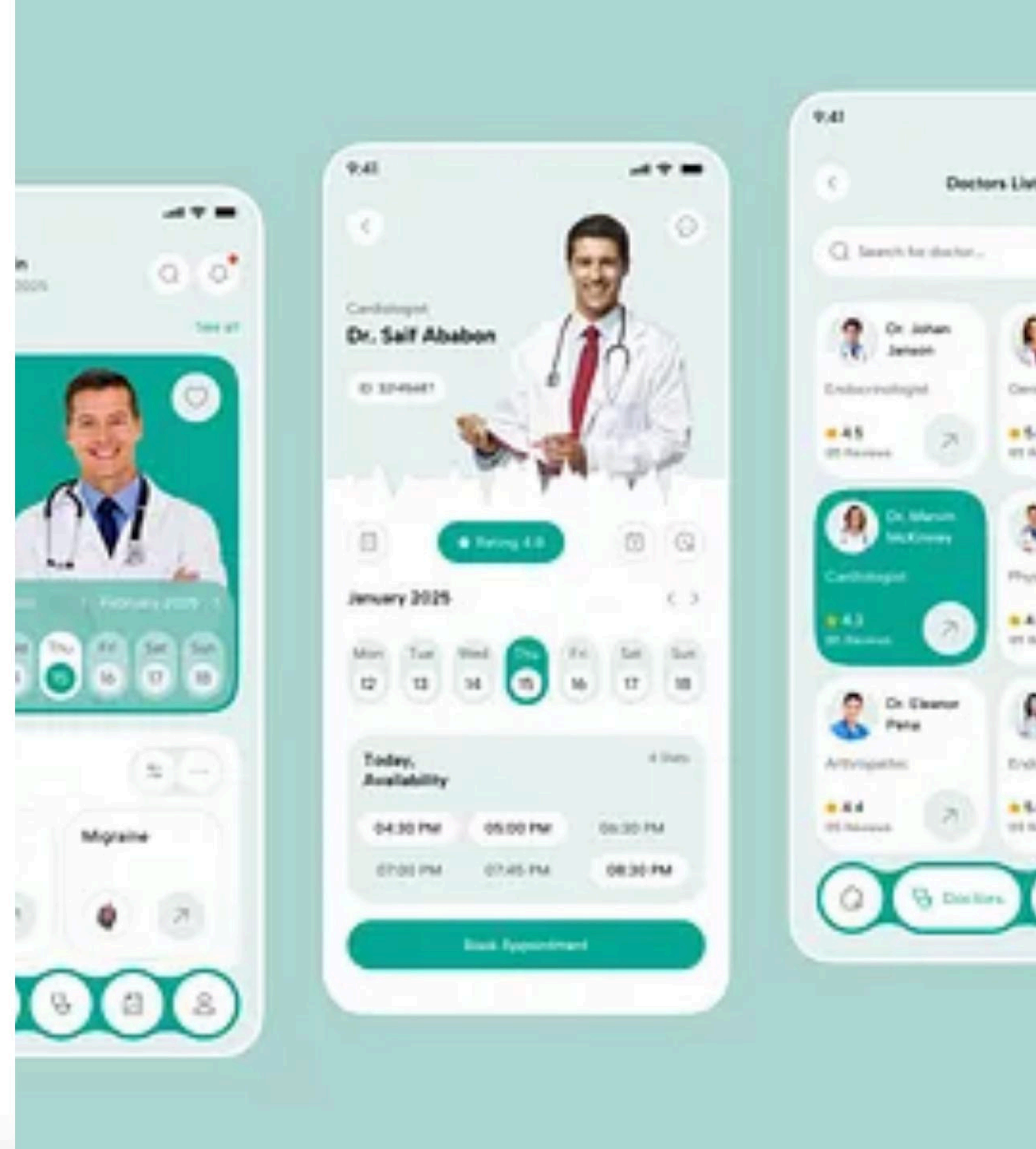
-  **Digitized Registration:** Automate token generation and patient intake procedures.
-  **Real-time Visibility:** Provide status updates and estimated waiting times for patients.
-  **History Management:** Store symptoms, prescriptions, and longitudinal visit history.
-  **Emergency Access:** Enable QR-code based quick retrieval of medical IDs.
-  **Data Security:** Ensure role-based access to sensitive medical information.

Our Proposed Solution

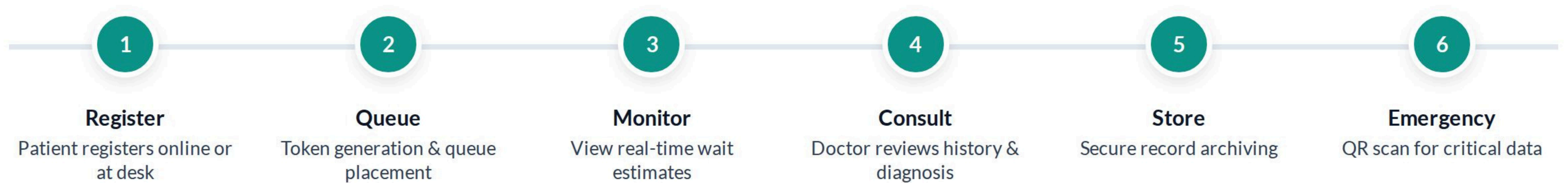
The system integrates OPD queue management with a secure digital patient history module.

Integrated Ecosystem: A unified platform for patients, doctors, and admin staff to interact without paper friction.

Patient interactions are securely encrypted and stored, while the Emergency QR module ensures life-saving data is always a scan away.



System Operational Workflow



Core System Features



Queue Tracking

Live monitoring of patient flow and wait times.



Smart EHR

Longitudinal history and e-prescriptions.



Emergency QR

Quick access to Medical IDs in accidents.



RBAC Security

Strict role-based data access control.

Technology Stack

Frontend

React.js / HTML5 / CSS3 / JavaScript for a responsive and intuitive user experience across devices.

Backend

Node.js with Express / Python Flask providing robust API services and business logic.

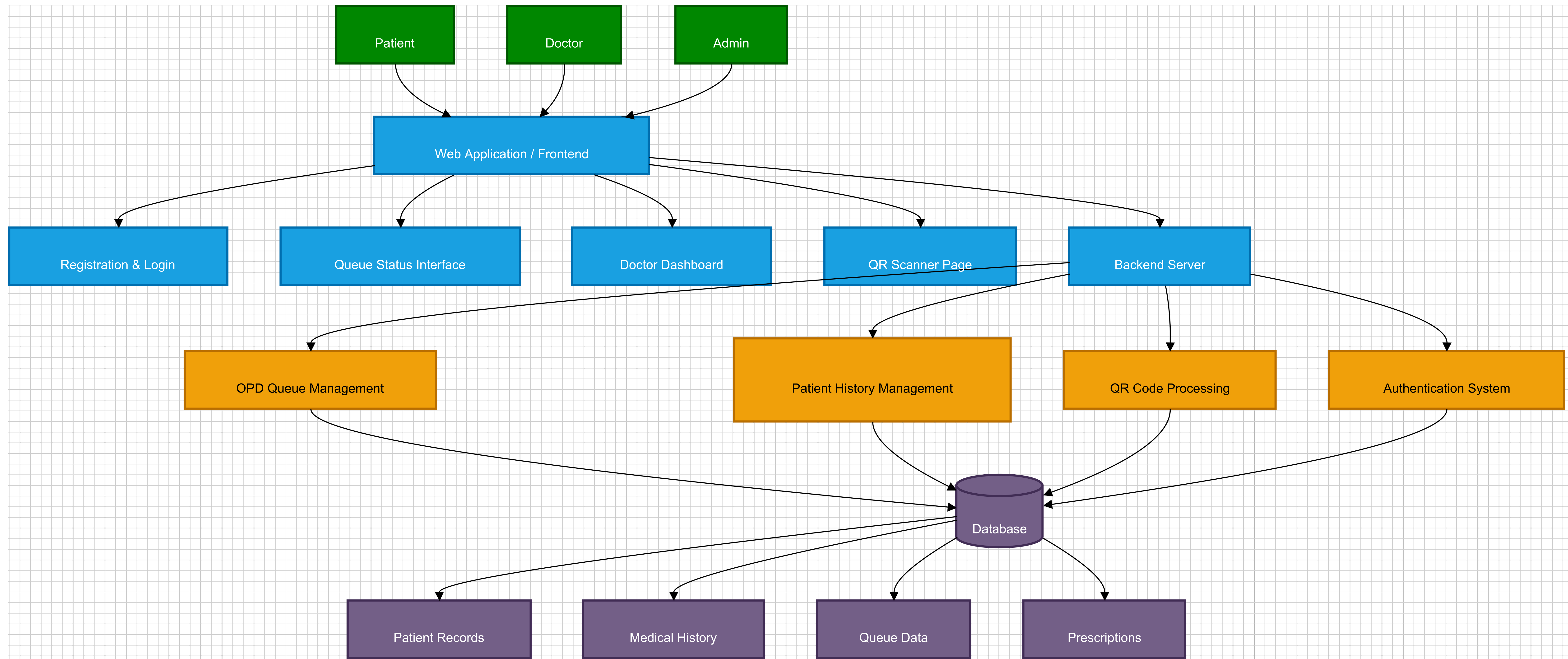
Database

MongoDB / PostgreSQL for high-performance storage of patient records and queue data.

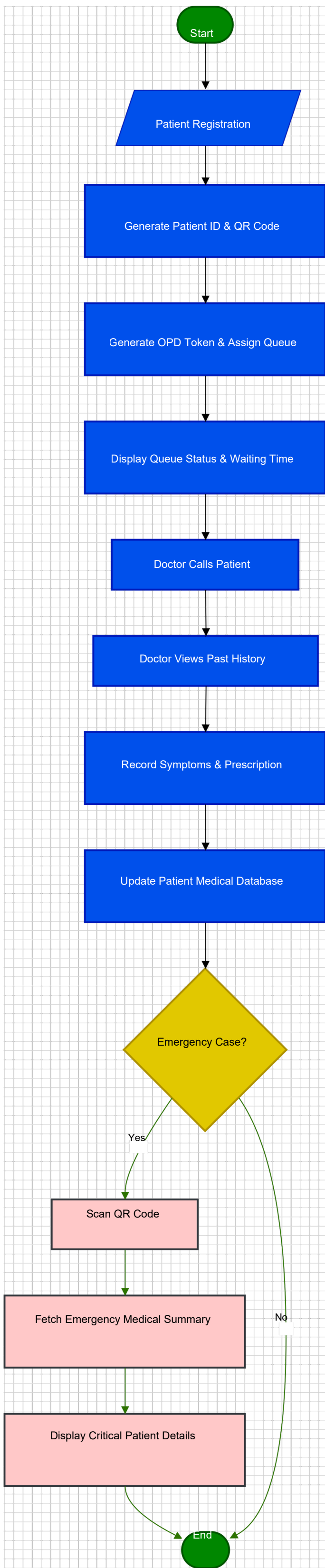
Tools & ML

Regression models for wait prediction, GitHub for version control, and Postman for testing.

Flowchart



Flowchart



| Scope & Conclusion

Target Application

Designed for small to medium hospitals and clinics. It serves as a decision-support and management tool.

Conclusion: This system is a practical, ethical solution for real-world healthcare, ensuring efficiency, patient safety, and better care delivery.



THANK YOU